ACML101

Physical Chemistry Assignment 1

Instructor

Prof. Shashank Deep

Fill in the blank

(i) At 300 K, the free energy of water vapour at 2 atm is _**smaller**_than water vapour at 5 atm. (greater/smaller)

(ii) At 1 atm, the free energy of water (liquid) at 300 K is **smaller** than water vapour at 300 K. (greater/smaller)

(iii) For $CO_2(g) \rightarrow CO_2(l)$, the sign of ΔG and ΔH at NTP are **positive** and **negative** respectively (positive/negative)

(iv) The sublimation temperature of dry ice will **increase** with increase in pressure (increase/decrease)

(v) The volume of 0.1 M H_2SO_4 required to neutralize 50 ml of 0.02 M NaOH is **5 mL**

(vi) For the reaction, $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$, ΔH= -120 kJ mol^{-1} and ΔS= -150 J K^{-1} mol^{-1}. The reaction will be spontaneous at temperature **smaller** than 800 K (greater/smaller)

(vii) The equilibrium constant for a reaction $A(g) + B(g) \rightleftharpoons C(g) + D(g)$ is 25. If 1 mol of A is mixed with 1 mol of B in a closed container, the number of mol of $A(g)$ at equilibrium will be **1/6**

(viii) For a first order decomposition reaction $2\ A(g) \rightarrow B(g) + C\ (s)$, the total pressure after 10 minute is found to be 300 Pa. If on completion of reaction, total pressure is observed to be 200 Pa, the rate constant for the reaction in min^{-1} is **0.0693 min^{-1}**